

Daniel Akinwale

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EDUCATION

Cornell University, College of Engineering, Ithaca NY

August 2022-May 2026

Bachelor of Science, Mechanical Engineering | M.Eng Mechanical & Aerospace Engineering

Relevant Courses:

CAD I & CAD II, MATLAB, Intro to Mechanical Design, Fluid Mechanics, Heat Transfer, System Dynamics, Mechatronics, Propulsion Aircraft & Rockets, Automotive Engineering, Electric-Drive Vehicle Engineering, Computer-Aided Manufacture, Dynamics of Flight Vehicles, Geometric Dimensioning & Tolerancing (GD&T), Additive Manufacturing, Design for Manufacture & Assembly (DFMA)

SPECIALIZED SKILLS

Tools: CNC (HAAS), Mill, Lathe, Band Saw, Waterjet Cutter, Laser Cutter, Benchtop CNC Mill

Programs: SolidWorks, Fusion360, AutoCAD, Microsoft 365: Excel/Word/PowerPoint, SQL, JavaScript, Java, Python, MATLAB, ANSYS, Arduino

Languages: Yoruba, English

PROFESSIONAL EXPERIENCE

Aerospace ADVERSARY Laboratory, Ithaca, NY, *AFRL Graduate Student Researcher*

March 2026-Present

- Assisting in the development of autonomous space infrastructure operations
- Working with system architecture, CAD and 3D printing, cube satellites, Arduino, lasers and whatever else is needed to create satellites for different laboratory objectives and missions

Emerson Manufacturing Lab, Ithaca NY, *MLS Crew Manager*

January 2024-Present

- Training students to use different metalworking machines in the manufacturing studio
- Teaching new trainers to use unfamiliar machines through instruction and designing training parts to test them
- Supervising people in shop by policing safety issues and showing proper methods of using tools
- Inventorying and replacing damaged and old equipment

GlobalFoundries, Malta NY, *Photolithography Equipment Engineering Intern*

June 2025-Aug 2025

- Supported in-house retrofits of various photolithography track tools from TEL and SEMES
- Modernized SEMES, ACT12, and ProV tools by installing cameras and DVRs in critical processing areas for faster detection of wafer defects
- Created solutions for tool panels in need of repair by designing cheaper replacement fastening method through CAD and manufacturing using 3D printers and magnets
- Gained overview of semiconductor manufacturing and photolithography processes in a Fab environment

SRW Engineering & Architecture, Philadelphia PA, *Mechanical Engineering Intern*

May 2024-Aug 2024

- Site visits of new, completed, and in progress buildings
- Creating AutoCAD floor plans for outlets, pipes, wiring, heaters, and other infrastructure
- Designing buildings based on client specifications and what they would be used for
 - Dividing large buildings into single family homes, maximizing space given max heights of floors and building itself
 - Incorporating client ideas like having buildings partially buried
- Learning about architecture, aesthetic considerations, and functional building design
- ASHRAE Training, OSHA 10-Hour Course

The VMC Group, Bloomington NJ, *Shock and Vibration Applications Engineering Intern*

June 2021-Aug 2021

- Worked with customers to provide technical insight for isolator selection in the aerospace and defense field.
- Designed, created, and optimized shock and vibration isolation systems to protect sensitive equipment against harsh environments like off-road terrain, turbulent flight, and bomb blasts.
- Learned to perform extensive modeling, simulation, and dynamic analysis of nonlinear six-degree-of-freedom (6DOF) system response to shock and vibration phenomena.

ORGANIZATIONS

Cornell Hyperloop (Project Team), *Composites Team Lead*

February 2023-Present

- Composites Team Lead in charge of designing carbon fiber aeroshell, designing manufacturing process team will follow, instructing team in process, and chemical safety enforcement
- Extensive CAD modeling, analysis, and machining for manufacture using SolidWorks, ANSYS, lathes, mills, and CNCing

National Society of Black Engineers (NSBE), *General Member*

October 2022-Present